PG No. 47 $T_{\mathrm{h}}(c)$ $m\bar{3}$ [cubic] (internal polar dipole)

* Harmonics for rank 0

$\vec{\mathbb{Q}}_{0}^{(1,1)}[q](A_g)$

** symmetry

1

** expression

$$\frac{\sqrt{3}Q_x x}{3}+\frac{\sqrt{3}Q_y y}{3}+\frac{\sqrt{3}Q_z z}{3}$$

* Harmonics for rank 1

$\vec{\mathbb{G}}_{1,1}^{(1,0)}[q](T_g)$, $\vec{\mathbb{G}}_{1,2}^{(1,0)}[q](T_g)$, $\vec{\mathbb{G}}_{1,3}^{(1,0)}[q](T_g)$

** symmetry

x

y

z

** expression

$$\frac{\sqrt{2}Q_y z}{2}-\frac{\sqrt{2}Q_z y}{2}$$

$$-\frac{\sqrt{2}Q_x z}{2}+\frac{\sqrt{2}Q_z x}{2}$$

$$\frac{\sqrt{2}Q_x y}{2}-\frac{\sqrt{2}Q_y x}{2}$$

$\vec{\mathbb{Q}}_{1,1}^{(1,-1)}[q](T_u)$, $\vec{\mathbb{Q}}_{1,2}^{(1,-1)}[q](T_u)$, $\vec{\mathbb{Q}}_{1,3}^{(1,-1)}[q](T_u)$

** symmetry

x

y

z

** expression

Q_x

Q_y

Q_z

$\vec{\mathbb{Q}}_{1,1}^{(1,1)}[q](T_u)$, $\vec{\mathbb{Q}}_{1,2}^{(1,1)}[q](T_u)$, $\vec{\mathbb{Q}}_{1,3}^{(1,1)}[q](T_u)$

** symmetry

x

y

z

** expression

$$\frac{\sqrt{10}Q_x\left(2x^2-y^2-z^2\right)}{10}+\frac{3\sqrt{10}Q_y xy}{10}+\frac{3\sqrt{10}Q_z xz}{10}$$

$$\frac{3\sqrt{10}Q_x xy}{10}-\frac{\sqrt{10}Q_y\left(x^2-2y^2+z^2\right)}{10}+\frac{3\sqrt{10}Q_z yz}{10}$$

$$\frac{3\sqrt{10}Q_x xz}{10}+\frac{3\sqrt{10}Q_y yz}{10}-\frac{\sqrt{10}Q_z\left(x^2+y^2-2z^2\right)}{10}$$

* Harmonics for rank 2

$\vec{\mathbb{Q}}_{2}^{(1,-1)}[q](E_g^{(a)})$

** symmetry

$$-\frac{\sqrt{2}x^2}{4}+\frac{\sqrt{6}ix^2}{4}-\frac{\sqrt{2}y^2}{4}-\frac{\sqrt{6}iy^2}{4}+\frac{\sqrt{2}z^2}{2}$$

** expression

$$-\frac{Q_x x\left(\sqrt{3}-3i\right)}{6}-\frac{Q_y y\left(\sqrt{3}+3i\right)}{6}+\frac{\sqrt{3}Q_z z}{3}$$

$\vec{\mathbb{Q}}_2^{(1,1)}[q](E_g^{(a)})$
** symmetry

$$-\frac{\sqrt{2}x^2}{4}+\frac{\sqrt{6}ix^2}{4}-\frac{\sqrt{2}y^2}{4}-\frac{\sqrt{6}iy^2}{4}+\frac{\sqrt{2}z^2}{2}$$

** expression

$$\frac{iQ_x x\left(3\sqrt{14}x^2+\sqrt{42}ix^2-7\sqrt{14}y^2+\sqrt{42}iy^2-2\sqrt{14}z^2-4\sqrt{42}iz^2\right)}{28}$$
$$+\frac{iQ_y y\left(7\sqrt{14}x^2+\sqrt{42}ix^2-3\sqrt{14}y^2+\sqrt{42}iy^2+2\sqrt{14}z^2-4\sqrt{42}iz^2\right)}{28}+\frac{iQ_z z\left(5\sqrt{14}x^2+3\sqrt{42}ix^2-5\sqrt{14}y^2+3\sqrt{42}iy^2-2\sqrt{42}iz^2\right)}{28}$$

$\vec{\mathbb{Q}}_2^{(1,-1)}[q](E_g^{(b)})$
** symmetry

$$-\frac{\sqrt{2}x^2}{4}-\frac{\sqrt{6}ix^2}{4}-\frac{\sqrt{2}y^2}{4}+\frac{\sqrt{6}iy^2}{4}+\frac{\sqrt{2}z^2}{2}$$

** expression

$$-\frac{Q_x x\left(\sqrt{3}+3i\right)}{6}-\frac{Q_y y\left(\sqrt{3}-3i\right)}{6}+\frac{\sqrt{3}Q_z z}{3}$$

$\vec{\mathbb{Q}}_2^{(1,1)}[q](E_g^{(b)})$
** symmetry

$$-\frac{\sqrt{2}x^2}{4}-\frac{\sqrt{6}ix^2}{4}-\frac{\sqrt{2}y^2}{4}+\frac{\sqrt{6}iy^2}{4}+\frac{\sqrt{2}z^2}{2}$$

** expression

$$-\frac{iQ_x x\left(3\sqrt{14}x^2-\sqrt{42}ix^2-7\sqrt{14}y^2-\sqrt{42}iy^2-2\sqrt{14}z^2+4\sqrt{42}iz^2\right)}{28}$$
$$-\frac{iQ_y y\left(7\sqrt{14}x^2-\sqrt{42}ix^2-3\sqrt{14}y^2-\sqrt{42}iy^2+2\sqrt{14}z^2+4\sqrt{42}iz^2\right)}{28}-\frac{iQ_z z\left(5\sqrt{14}x^2-3\sqrt{42}ix^2-5\sqrt{14}y^2-3\sqrt{42}iy^2+2\sqrt{42}iz^2\right)}{28}$$

$\vec{\mathbb{Q}}_{2,1}^{(1,-1)}[q](T_g)$, $\vec{\mathbb{Q}}_{2,2}^{(1,-1)}[q](T_g)$, $\vec{\mathbb{Q}}_{2,3}^{(1,-1)}[q](T_g)$
** symmetry

$$\sqrt{3}yz$$

$$\sqrt{3}xz$$

$$\sqrt{3}xy$$

** expression

$$\frac{\sqrt{2}Q_y z}{2}+\frac{\sqrt{2}Q_z y}{2}$$

$$\frac{\sqrt{2}Q_x z}{2}+\frac{\sqrt{2}Q_z x}{2}$$

$$\frac{\sqrt{2}Q_x y}{2}+\frac{\sqrt{2}Q_y x}{2}$$

$\vec{\mathbb{Q}}_{2,1}^{(1,1)}[q](T_g)$, $\vec{\mathbb{Q}}_{2,2}^{(1,1)}[q](T_g)$, $\vec{\mathbb{Q}}_{2,3}^{(1,1)}[q](T_g)$
** symmetry

$$\sqrt{3}yz$$

$$\sqrt{3}xz$$

$$\sqrt{3}xy$$

** expression

$$\frac{5\sqrt{7}Q_x xyz}{7}-\frac{\sqrt{7}Q_y z\left(x^2-4y^2+z^2\right)}{7}-\frac{\sqrt{7}Q_z y\left(x^2+y^2-4z^2\right)}{7}$$

$$\frac{\sqrt{7}Q_x z\left(4x^2-y^2-z^2\right)}{7}+\frac{5\sqrt{7}Q_y xyz}{7}-\frac{\sqrt{7}Q_z x\left(x^2+y^2-4z^2\right)}{7}$$

$$\frac{\sqrt{7}Q_x y\left(4x^2-y^2-z^2\right)}{7}-\frac{\sqrt{7}Q_y x\left(x^2-4y^2+z^2\right)}{7}+\frac{5\sqrt{7}Q_z xyz}{7}$$

$\vec{\mathbb{G}}_2^{(1,0)}[q](E_u^{(a)})$

** symmetry

$$-\frac{\sqrt{2}x^2}{4}+\frac{\sqrt{6}ix^2}{4}-\frac{\sqrt{2}y^2}{4}-\frac{\sqrt{6}iy^2}{4}+\frac{\sqrt{2}z^2}{2}$$

** expression

$$\frac{Q_x yz\left(\sqrt{3}+i\right)}{2}-\frac{Q_y xz\left(\sqrt{3}-i\right)}{2}-iQ_z xy$$

$\vec{\mathbb{G}}_2^{(1,0)}[q](E_u^{(b)})$

** symmetry

$$-\frac{\sqrt{2}x^2}{4}-\frac{\sqrt{6}ix^2}{4}-\frac{\sqrt{2}y^2}{4}+\frac{\sqrt{6}iy^2}{4}+\frac{\sqrt{2}z^2}{2}$$

** expression

$$\frac{Q_x yz\left(\sqrt{3}-i\right)}{2}-\frac{Q_y xz\left(\sqrt{3}+i\right)}{2}+iQ_z xy$$

$\vec{\mathbb{G}}_{2,1}^{(1,0)}[q](T_u)$, $\vec{\mathbb{G}}_{2,2}^{(1,0)}[q](T_u)$, $\vec{\mathbb{G}}_{2,3}^{(1,0)}[q](T_u)$

** symmetry

$$\sqrt{3}yz$$

$$\sqrt{3}xz$$

$$\sqrt{3}xy$$

** expression

$$\frac{\sqrt{2}Q_x\left(y-z\right)\left(y+z\right)}{2}-\frac{\sqrt{2}Q_y xy}{2}+\frac{\sqrt{2}Q_z xz}{2}$$

$$\frac{\sqrt{2}Q_x xy}{2}-\frac{\sqrt{2}Q_y\left(x-z\right)\left(x+z\right)}{2}-\frac{\sqrt{2}Q_z yz}{2}$$

$$-\frac{\sqrt{2}Q_x xz}{2}+\frac{\sqrt{2}Q_y yz}{2}+\frac{\sqrt{2}Q_z\left(x-y\right)\left(x+y\right)}{2}$$

* Harmonics for rank 3

$\vec{\mathbb{G}}_3^{(1,0)}[q](A_g)$

** symmetry

$$\sqrt{15}xyz$$

** expression

$$\frac{\sqrt{5}Q_x x\left(y-z\right)\left(y+z\right)}{2}-\frac{\sqrt{5}Q_y y\left(x-z\right)\left(x+z\right)}{2}+\frac{\sqrt{5}Q_z z\left(x-y\right)\left(x+y\right)}{2}$$

$\vec{\mathbb{G}}_{3,1}^{(1,0)}[q](T_g,1)$, $\vec{\mathbb{G}}_{3,2}^{(1,0)}[q](T_g,1)$, $\vec{\mathbb{G}}_{3,3}^{(1,0)}[q](T_g,1)$

** symmetry

$$x^3-\frac{3xy^2}{2}-\frac{3xz^2}{2}$$

$$-\frac{3x^2y}{2}+y^3-\frac{3yz^2}{2}$$

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$\frac{\sqrt{3}Q_y z\left(4x^2-y^2-z^2\right)}{4}-\frac{\sqrt{3}Q_z y\left(4x^2-y^2-z^2\right)}{4}$$

$$\frac{\sqrt{3}Q_x z\left(x^2-4y^2+z^2\right)}{4}-\frac{\sqrt{3}Q_z x\left(x^2-4y^2+z^2\right)}{4}$$

$$-\frac{\sqrt{3}Q_x y\left(x^2+y^2-4z^2\right)}{4}+\frac{\sqrt{3}Q_y x\left(x^2+y^2-4z^2\right)}{4}$$

$\vec{\mathbb{G}}_{3,1}^{(1,0)}[q](T_g,2)$, $\vec{\mathbb{G}}_{3,2}^{(1,0)}[q](T_g,2)$, $\vec{\mathbb{G}}_{3,3}^{(1,0)}[q](T_g,2)$
** symmetry

$$\frac{\sqrt{15}xy^2}{2}-\frac{\sqrt{15}xz^2}{2}$$

$$-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$$

$$\frac{\sqrt{15}x^2z}{2}-\frac{\sqrt{15}y^2z}{2}$$

** expression

$$-\sqrt{5}Q_x xyz+\frac{\sqrt{5}Q_y z\left(2x^2+y^2-z^2\right)}{4}+\frac{\sqrt{5}Q_z y\left(2x^2-y^2+z^2\right)}{4}$$

$$\frac{\sqrt{5}Q_x z\left(x^2+2y^2-z^2\right)}{4}-\sqrt{5}Q_y xyz-\frac{\sqrt{5}Q_z x\left(x^2-2y^2-z^2\right)}{4}$$

$$\frac{\sqrt{5}Q_x y\left(x^2-y^2+2z^2\right)}{4}-\frac{\sqrt{5}Q_y x\left(x^2-y^2-2z^2\right)}{4}-\sqrt{5}Q_z xyz$$

$\vec{\mathbb{Q}}_{3}^{(1,-1)}[q](A_u)$
** symmetry

$$\sqrt{15}xyz$$

** expression

$$Q_x yz+Q_y xz+Q_z xy$$

$\vec{\mathbb{Q}}_{3}^{(1,1)}[q](A_u)$
** symmetry

$$\sqrt{15}xyz$$

** expression

$$\frac{\sqrt{15}Q_x yz\left(6x^2-y^2-z^2\right)}{6}-\frac{\sqrt{15}Q_y xz\left(x^2-6y^2+z^2\right)}{6}-\frac{\sqrt{15}Q_z xy\left(x^2+y^2-6z^2\right)}{6}$$

$\vec{\mathbb{Q}}_{3,1}^{(1,-1)}[q](T_u,1)$, $\vec{\mathbb{Q}}_{3,2}^{(1,-1)}[q](T_u,1)$, $\vec{\mathbb{Q}}_{3,3}^{(1,-1)}[q](T_u,1)$
** symmetry

$$x^3-\frac{3xy^2}{2}-\frac{3xz^2}{2}$$

$$-\frac{3x^2y}{2}+y^3-\frac{3yz^2}{2}$$

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$\frac{\sqrt{15}Q_x\left(2x^2-y^2-z^2\right)}{10}-\frac{\sqrt{15}Q_y xy}{5}-\frac{\sqrt{15}Q_z xz}{5}$$

$$-\frac{\sqrt{15}Q_x xy}{5}-\frac{\sqrt{15}Q_y\left(x^2-2y^2+z^2\right)}{10}-\frac{\sqrt{15}Q_z yz}{5}$$

$$-\frac{\sqrt{15}Q_x xz}{5}-\frac{\sqrt{15}Q_y yz}{5}-\frac{\sqrt{15}Q_z\left(x^2+y^2-2z^2\right)}{10}$$

$\vec{\mathbb{Q}}_{3,1}^{(1,-1)}[q](T_u,2)$, $\vec{\mathbb{Q}}_{3,2}^{(1,-1)}[q](T_u,2)$, $\vec{\mathbb{Q}}_{3,3}^{(1,-1)}[q](T_u,2)$

** symmetry

$$\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$$

$$-\frac{\sqrt{15}x^2y}{2} + \frac{\sqrt{15}yz^2}{2}$$

$$\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$$

** expression

$$\frac{Q_x\left(y-z\right)\left(y+z\right)}{2} + Q_yxy - Q_zxz$$

$$-Q_xxy - \frac{Q_y\left(x-z\right)\left(x+z\right)}{2} + Q_zyz$$

$$Q_xxz - Q_yyz + \frac{Q_z\left(x-y\right)\left(x+y\right)}{2}$$

$\vec{\mathbb{Q}}_{3,1}^{(1,1)}[q](T_u, 1)$, $\vec{\mathbb{Q}}_{3,2}^{(1,1)}[q](T_u, 1)$, $\vec{\mathbb{Q}}_{3,3}^{(1,1)}[q](T_u, 1)$

** symmetry

$$x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$$

$$-\frac{3x^2y}{2} + y^3 - \frac{3yz^2}{2}$$

$$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$$

** expression

$$\frac{Q_x\left(8x^4 - 24x^2y^2 - 24x^2z^2 + 3y^4 + 6y^2z^2 + 3z^4\right)}{12} + \frac{5Q_yxy\left(4x^2 - 3y^2 - 3z^2\right)}{12} + \frac{5Q_zxz\left(4x^2 - 3y^2 - 3z^2\right)}{12}$$

$$-\frac{5Q_xxy\left(3x^2 - 4y^2 + 3z^2\right)}{12} + \frac{Q_y\left(3x^4 - 24x^2y^2 + 6x^2z^2 + 8y^4 - 24y^2z^2 + 3z^4\right)}{12} - \frac{5Q_zyz\left(3x^2 - 4y^2 + 3z^2\right)}{12}$$

$$-\frac{5Q_xxz\left(3x^2 + 3y^2 - 4z^2\right)}{12} - \frac{5Q_yyz\left(3x^2 + 3y^2 - 4z^2\right)}{12} + \frac{Q_z\left(3x^4 + 6x^2y^2 - 24x^2z^2 + 3y^4 - 24y^2z^2 + 8z^4\right)}{12}$$

$\vec{\mathbb{Q}}_{3,1}^{(1,1)}[q](T_u, 2)$, $\vec{\mathbb{Q}}_{3,2}^{(1,1)}[q](T_u, 2)$, $\vec{\mathbb{Q}}_{3,3}^{(1,1)}[q](T_u, 2)$

** symmetry

$$\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$$

$$-\frac{\sqrt{15}x^2y}{2} + \frac{\sqrt{15}yz^2}{2}$$

$$\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$$

** expression

$$\frac{\sqrt{15}Q_x\left(y-z\right)\left(y+z\right)\left(6x^2 - y^2 - z^2\right)}{12} - \frac{\sqrt{15}Q_yxy\left(2x^2 - 5y^2 + 9z^2\right)}{12} + \frac{\sqrt{15}Q_zxz\left(2x^2 + 9y^2 - 5z^2\right)}{12}$$

$$-\frac{\sqrt{15}Q_xxy\left(5x^2 - 2y^2 - 9z^2\right)}{12} + \frac{\sqrt{15}Q_y\left(x-z\right)\left(x+z\right)\left(x^2 - 6y^2 + z^2\right)}{12} - \frac{\sqrt{15}Q_zyz\left(9x^2 + 2y^2 - 5z^2\right)}{12}$$

$$\frac{\sqrt{15}Q_xxz\left(5x^2 - 9y^2 - 2z^2\right)}{12} + \frac{\sqrt{15}Q_yyz\left(9x^2 - 5y^2 + 2z^2\right)}{12} - \frac{\sqrt{15}Q_z\left(x-y\right)\left(x+y\right)\left(x^2 + y^2 - 6z^2\right)}{12}$$